

UNIVERSITY of WASHINGTON

THE CHOICE INSTITUTE

School of Pharmacy

ANNUAL REPORT 2024

THE COMPARATIVE HEALTH OUTCOMES, POLICY & ECONOMICS INSTITUTE

TABLE OF CONTENTS

3 MISSION & DIVERSITY STATEMENT

4 FACULTY AND STAFF

6 DIRECTOR'S WELCOME

7 CLASS OF 2024

9 NEW STUDENTS AND FELLOWS

12 CURRENT STUDENTS

14 INSTITUTE UPDATES

25 RESEARCH HIGHLIGHTS

33 ACTIVITIES AND ACCOMPLISHMENTS

42 ALUMNI HIGHLIGHTS

44 PUBLICATIONS

47 DONOR IMPACT

OUR MISSION

Research: Develop innovative methods and generate actionable evidence about the effectiveness, safety, and value of medical products and services that improve decision making in health care and policy.

Training: Provide transformative training in health economics, outcomes research, and policy analysis to meet contemporary health care challenges regionally, nationally, and globally.

Service: Be a leading resource on the effectiveness, safety, and value of medical products and services by providing expertise and facilitating collaborations at the regional, national, and global level.



CHOICE STATEMENT ON DIVERSITY, EQUITY, AND INCLUSION

CHOICE is committed to creating an inclusive academic community where every individual is treated fairly and with dignity and respect. We strongly believe that diversity enriches learning, broadens the perspectives of all in our program, and improves our sense of community. Diversity requires an atmosphere of inclusion, tolerance, respect, and equity.

Our equality and diversity principles:

- We value the tremendous diversity of the human experience and believe that this diversity strengthens our communities and our program.
- We believe that discrimination or exclusion based on individual characteristics and circumstances, such as age; disability; caring or dependency responsibilities; gender or gender identity; marriage and civil partnership status; political opinion; pregnancy and maternity; race, color, caste, nationality, ethnic or national origin; religion or belief; sexual orientation; socio-economic status; or other distinctions, is unjust and represents a waste of talent and a denial of opportunity for self-fulfillment.
- We recognize that patterns of under-representation and differences in outcomes at CHOICE can be challenged through positive action.
- We respect the rights of individuals, including the right to hold their own opinions and beliefs, but will not allow these opinions to be manifested in a way that is hostile or degrading to others.
- We expect commitment and involvement from all our faculty, staff, and students in working towards the achievement of our vision.

CHOICE FACULTY & STAFF

Core Faculty

Jennifer Bacci, MPH, PharmD
Associate Professor, Endowed Professor in Innovative Pharmacy Practice (I2P2)

Aasthaa Bansal, PhD
Associate Professor

Douglas Barthold, PhD
Research Associate Professor, CHOICE Endowed Faculty Fellow

Anirban Basu, PhD
Stergachis Family Endowed Director & Professor

Josh Carlson, MPH, PhD
Professor, Director, Graduate Program

Clayton English, PharmD
Assistant Professor

Shelly Gray, PharmD, MS
Plein Endowed Professor for the Director of the Center for Geriatric Pharmacy, Research, Education and Outreach

Ryan Hansen, PharmD, PhD
Professor, Interim Chair, Department of Pharmacy

Laura Hart, PharmD, MS
Assistant Professor

Noémi Kreif, PhD, MA, Msc
Assistant Professor

Kyueun Lee, PhD
Assistant Professor of Decision Sciences

Jing Li, MA, PhD
Associate Professor of Health Economics

Andy Stergachis, PhD
Professor

Sean Sullivan, PhD
Professor

David Veenstra, PharmD, PhD
Professor

Emeritus Faculty

Beth Devine, PharmD, MBA, PhD
Professor Emeritus

Louis Garrison Jr., PhD
Professor Emeritus

Thomas Hazlet, DrPH, PharmD
Associate Professor Emeritus

Adjunct Faculty

David Au, MS, MD
Adjunct Professor

Brian Bresnahan, PhD
Adjunct Research Associate Professor

Todd Edwards, PhD
Adjunct Professor

David Flum, MD, MPH
Adjunct Professor

Lurdes Inoue, MS, MS, PhD
Adjunct Professor

Jerry Jarvik, MD, MPH
Adjunct Professor

Larry Kessler, ScD
Adjunct Professor

Gary Lyman, MD, MPH
Adjunct Professor

Scott Ramsey, MD, PhD
Adjunct Professor, Director, Hutchinson Institute for Cancer Outcomes Research and Evaluation

Staff

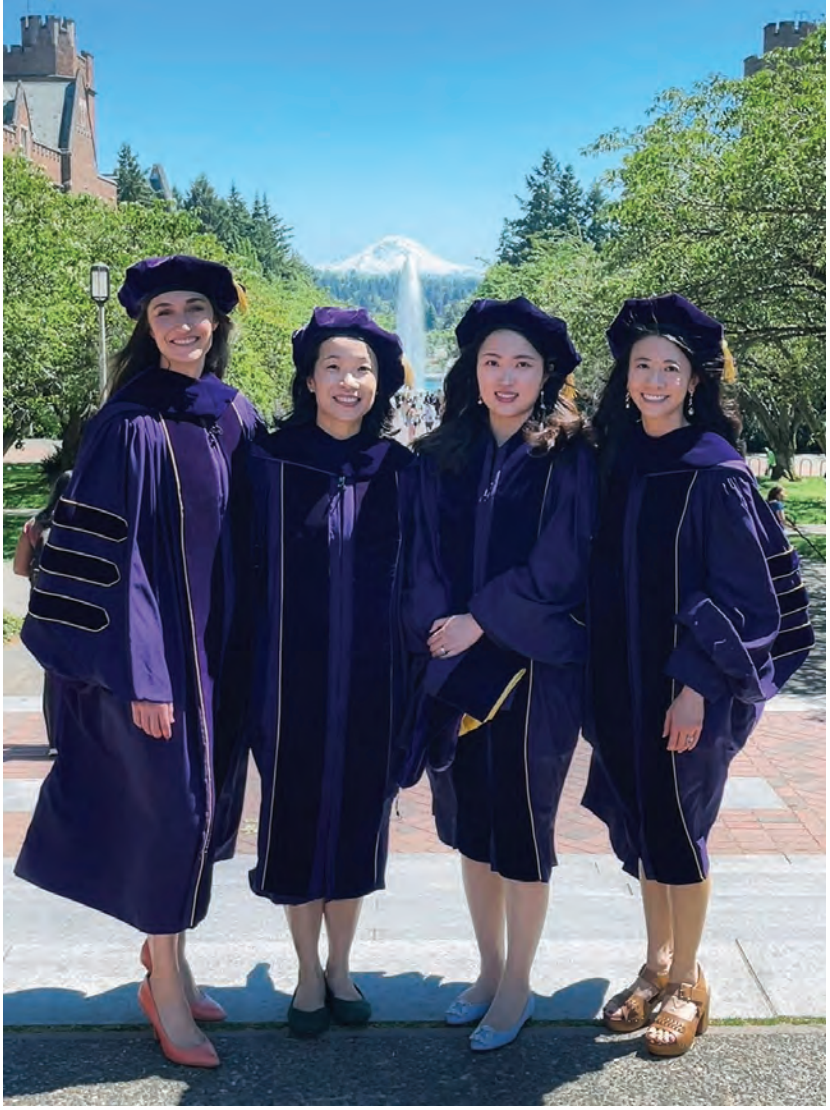
Leana de la Torre
Online Certificate Program Manager

Marina Gano, M.Ed.
Graduate Program & Operations Manager

Greg Guzauskas, PhD
Senior Research Scientist

Linda Luu, MS
Biostatistician

OPPOSITE PAGE — TOP LEFT: RACHEL WITTENAUER, SARA KHOR, YILIN CHEN, AND JACINDA TRAN CELEBRATING THEIR GRADUATION IN FRONT OF RAINIER VISTA; TOP RIGHT: THE 2024 CHOICE PHD GRADUATES AND FACULTY; BOTTOM: THE CHOICE FACULTY AND STAFF AT THE 2024 ANNUAL RETREAT







DIRECTOR'S WELCOME



Dear Friends and Colleagues,

The 2023–2024 Annual Report highlights the continued excellence of the CHOICE Institute. ScholarGPS recently recognized the School of Pharmacy at the University of Washington as the world's third highest-ranked Institute in pharmacy and pharmaceutical sciences based on its lifetime productivity. Multiple CHOICE faculty members were recognized as leading contributors in the fields of Health Economics and Managed Care. Congratulations to Professor Andy Stergachis for being named a National Academy of Medicine (NAM) Fellow, Associate Professor Jenni Bacci for being named an American College of Clinical Pharmacy (ACCP) Fellow, and Research Associate Professor Douglas Barthold for being appointed by Governor Jay Inslee to serve on the Washington State Prescription Drug Affordability Board (PDAB). We are also excited to welcome the newest faculty members at CHOICE, Assistant Professors Noémi Kreif and Laura Hart.

Last fall, we hosted our Annual Symposium, "Artificial Intelligence, Real Consequences: The Side Effects of AI in Healthcare." Moderated by Associate Professor Aasthaa Bansal, the event featured insightful discussions on the potential benefits and unintended consequences of AI technologies in medical practice, health economics, and outcomes research. The symposium highlighted the critical need for ethical AI deployment in healthcare, the potential benefits of advanced AI models like LLMs tempered by unique risks, and the role of rigorous research methodologies in ensuring AI-driven decisions are reliable and equitable. Several institute members are working closely on these topics. We are also launching a new certificate in Data Sciences in HEOR this Fall.

A round of kudos applies to many of our students and alums. I highlight a few here. PhD student Ashley Tabah was honored with the School of Pharmacy Graduate Student Leadership Award. PhD student Felipe Montano-Campos was selected to be a Fellow of the Latino Center for Health, an interdisciplinary health sciences research center at the University of Washington committed to improving the health and well-being of Latinx individuals, families, and communities in Washington State. Sara Khor, PhD ('24) was awarded the American Society of Clinical Oncology (ASCO) Conquer Cancer Merit Award for her team's study that examined the association between pre-diagnosis financial strain and later-stage cancer diagnosis. Rachel Wittenauer, MPH, PhD ('24), was selected for the 2024 School of Pharmacy Outstanding Dissertation Award.

Alum Jonathan Campbell, MS, PhD ('07) received the 2024 PAA Distinguished Alumni - Pharmaceutical Sciences/Research Award. Alum Meng Li, PhD ('18) received the First Prize award of the IVI Valuing Innovation Project (VIP) for her paper co-authored with Professor Emeritus Lou Garrison. We also welcome our new cohort of MS fellows joining us this July and the new cohort of PhD students entering in Fall. Finally, congrats to our graduating PhD students: Joyce Jiang, Sara Khor, Jacinda Tran, and Rachel Wittenauer, and our graduating MS students: Emma Behan, Jennifer Deem, Jake Earl, Filmon Haile, Alexandra Miller, and Richard Ta.

I am thankful to the entire Department of Pharmacy staff, especially Marina Gano and Leana de la Torre, for their wide-ranging contribution to the Institute's administration and keeping the engine running. I am also thankful to all our colleagues in the HEOR field who have supported us through myriad channels.

We are excited to host the Western Pharmacoeconomics Conference (WPEC) in Seattle from March 31 – April 2, 2025. I look forward to seeing many of our friends and colleagues there and getting to know the trainees from our peer institutions. I hope you will consider supporting us in this endeavor.

I wish you all a healthy 2024–2025 year ahead.

Sincerely,



Anirban Basu
Stergachis Family Endowed Director and Professor
The CHOICE Institute

CLASS OF 2024

DOCTOR OF PHILOSOPHY (HEALTH ECONOMICS & OUTCOMES RESEARCH)



Jacinda Tran Fulcher

MBA 2015, University of Baltimore; PharmD 2014, University of Maryland; BS 2009, University of California, Los Angeles

DISSERTATION: *Understanding HIV pre-exposure prophylaxis use in the United States and the potential impact of community pharmacies*

CHAIR: Ryan Hansen

CURRENT POSITION: Research Scientist, Bill & Melinda Gates Foundation, Institute for Disease Modeling



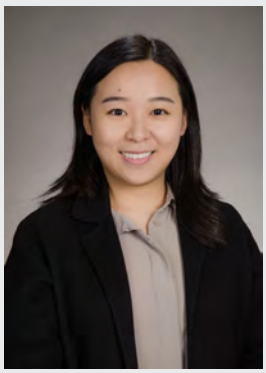
Sara Khor

MSc 2008, University of Toronto; BS 2005, University of Toronto

DISSERTATION: *Incorporating Equity into Healthcare Decision Making Around New Technologies*

CHAIR: Aasthaa Bansal & Josh Carlson

CURRENT POSITION: Research Scholar, Department of Health Policy at Stanford University



Shangqing Joyce Jiang

MPH 2019, Columbia University; BS 2015, Peking University, China

DISSERTATION: *Risk Prediction and Value of Polygenic Risk Scores in Colorectal Cancer Screening*

CHAIR: Dave Veenstra



Rachel Wittenauer

MPH 2020, University of Washington; BA 2013, Johns Hopkins University

DISSERTATION: *Equitable access and reimbursement for pharmacy-based services: A case study of adult vaccinations*

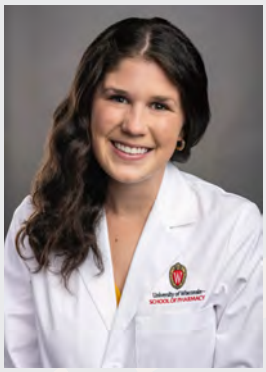
CHAIR: Andy Stergachis

CURRENT POSITION: Postdoctoral Scientist in Health Economics, World Health Organization, Lyon France





MASTER OF SCIENCE (HEALTH ECONOMICS & OUTCOMES RESEARCH)



Emma Behan

PharmD 2023, University of Wisconsin-Madison; BS 2020, University of Wisconsin-Madison

THESIS: *Healthcare resource utilization and costs of Medicare-enrolled patients with HR+/HER2- metastatic breast cancer treated with CDK4/6i in the first line setting*

CHAIR: Aasthaa Bansal & Dave Veenstra

CURRENT POSITION: Industry Fellowship Year 2; Genentech, San Francisco, CA



Jennifer Deem

PhD 2016, University of Washington; BS 2007, University of Washington

THESIS: *Real world treatment patterns and outcomes among patients with early non-small cell lung cancer*

CHAIR: Josh Carlson

CURRENT POSITION: Industry Fellowship Year 2; Pfizer, Seattle, WA



Jake Earl

PharmD 2023, University of Utah

THESIS: *Identifying and Characterizing Commercially Insured HFpEF Patients with Low vs. High Healthcare Resource Utilization*

CHAIR: Ryan Hansen

CURRENT POSITION: Industry Fellowship Year 2; Bayer, Whippany, NJ



Filmon Haile

PharmD 2023, University of Southern California; BS 2018, University of California, Los Angeles

THESIS: *Treatment Patterns and Patient Characteristics in Misdiagnosis of Bipolar Disorder-I Disorder*

CHAIR: Kyu Lee

CURRENT POSITION: Industry Fellowship Year 2; Abbvie, Florham Park, NJ



Alexandra Miller

MPH 2023, University of Texas; PharmD 2023, University of Texas at Austin; BA 2019, Wake Forest University

THESIS: *Exploring Demographic, Geographical, and Clinical Factors Associated with Persistence to High-Efficacy Therapies in Multiple Sclerosis*

CHAIR: Aasthaa Bansal & Dave Veenstra

CURRENT POSITION: Industry Fellowship Year 2; Genentech, San Francisco, CA



Richard Ta

PharmD 2023, University of Southern California; BS 2018, University of California-Davis

THESIS: *Targeted Therapies and Healthcare Resource Utilization and Cost Among Patients with Atopic Dermatitis*

CHAIR: Jing Li

CURRENT POSITION: Industry Fellowship Year 2; AbbVie, Mettawa, IL

WELCOME: NEW STUDENTS AND FELLOWS

PHD STUDENTS

Preetika Banerjee



MSPH, Global Disease Epidemiology and Control, Johns Hopkins University; Bachelor of Engineering in Biotechnology, PES Institute of Technology, Bangalore, India

Preetika joins us most recently from the Fred Hutch Cancer Center, where she worked as a Data Coordinator Lead, focusing on projects looking at pharmacy-based models of PrEP and PEP delivery in Kenya. Preetika holds an MSPH in Global Disease Epidemiology and Control from Johns Hopkins Bloomberg School of Public Health and a Bachelor of Engineering in Biotechnology from PES Institute of Technology in Bangalore, India. Her research interests include infectious diseases, preventative interventions, disease modeling, health equity, and access to care.

Thomas Hopkins



MS, Pharmaceutical Outcomes Research & Policy, University of Washington; PharmD, University of Illinois Chicago; BS, Biological Sciences, Loyola University Chicago

Tom is an incoming first-year PhD student at the CHOICE Institute. As a pharmacist with six years of experience in HEOR across academia and the pharmaceutical industry, he brings a wealth of expertise to his studies. Tom's interests include economic modeling, predictive modeling with machine learning, and incorporating social determinants of health into research design. He has worked in various therapeutic areas, including Alzheimer's disease, oncology, and women's health. Currently, Tom is focused on applying these methodologies to the fields of addiction, alcoholism, and mental health.

Fangzhe Sam Han



MSPH in Global Disease Epidemiology and Control, Johns Hopkins Bloomberg School of Public Health; BA, Mathematics, New York University

Sam recently graduated from Johns Hopkins University Bloomberg School of Public Health with a MSPH degree in Global Disease and Epidemiology. Prior to that, he obtained a Bachelor of Arts degree in mathematics at New York University. He has worked in the life sciences and pharmaceutical consulting space during several of his internships, where he found his passion for HEOR. He is passionate about finding better ways to help assess the value of pharmaceutics and medical devices. He is really looking forward to working and learning from the renowned faculty members at the CHOICE Institute.

Ruihan Helen Qin



MS in Biostatistics (Data Science Pathway), Yale University; Bachelor of Economics (International Trade and Economics), Central University of Finance and Economics, Beijing, China

Ruihan obtained her master's degree in biostatistics from Yale University and her bachelor's degree in economics from the Central University of Finance and Economics in China. Before joining UW, she worked as a statistician at the Yale/YNHH Center for Outcomes Research & Evaluation, where she focused on developing outcome measures to evaluate hospital performance. Ruihan's research interests include outcomes research and predictive modeling. She is very excited to join UW CHOICE and engage in research projects under the guidance of distinguished faculty members.

MS FELLOWS

Minseon Chung



UW/Novo Nordisk HEOR Fellow 2024–2026

PharmD & MPH, The State University of New York at Buffalo; BS Biology, Stony Brook University

Minseon recently graduated with a PharmD and MPH from the University at Buffalo. She cultivated her interests in health economics and real-world evidence through research projects analyzing national cohort data of pregnant women, identifying associations between socioeconomic factors and perinatal substance use. Her passion for HEOR grew stronger during her rotations, where she witnessed the impact of value assessment in clinical decision-making. Minseon's current research interests include economic evaluation and modeling. She looks forward to expanding her technical skillset and gaining experience in the field while learning from leading experts at the CHOICE Institute and Novo Nordisk.

Rachel Kneitel



UW/AbbVie HEOR Fellow 2024–2026

PharmD, St. John's University, New York

Rachel is a recent PharmD graduate from St. John's University. During pharmacy school, her industry internship and rotations in HEOR, Market Access Analytics, and pharmacoeconomic research on antibody-drug conjugates shaped her interest in the field. These experiences have allowed her to design compelling value stories by linking real-world data to draw valuable insights about healthcare practices and strive to capture the patient journey. Her current research interests include real-world evidence generation, cost-eff ctiveness modeling, and patient-reported outcomes. She is excited to develop her HEOR skillsets further and work closely with experts at the CHOICE Institute and AbbVie.

Divya Jain



UW/Genentech HEOR Fellow 2024–2026

PharmD, University of Colorado; BS Biochemistry, University of California, Los Angeles

Divya is an incoming first-year UW-Genentech CHOICE fellow and a recent PharmD graduate specializing in HEOR. With a diverse array of experiences, Divya is motivated to improve patient access and outcomes, with a particular focus on Multiple Sclerosis. In her free time, she enjoys comedy and action movies, sci-fi nd fantasy books, and exploring zoos, aquariums, and museums. Divya looks forward to her time at the CHOICE Institute and new adventures in Seattle!

Jean Lee



UW/Bayer HEOR Fellow 2024–2026

PharmD, University of Illinois at Chicago; BA Economics, University of Michigan, Ann Arbor

Jean recently graduated from the University of Illinois at Chicago School of Pharmacy. She developed her interest in HEOR and the pharmaceutical industry by participating in HEOR research, completing rotations at BMS and Cencora, and obtaining a pharmacoepidemiology certificate during pharmacy school. Jean looks forward to learning unique techniques applicable to HEOR, such as economic modeling and evidence generation, by working closely with the renowned experts at the UW CHOICE Institute. She aims to utilize and apply this education and experience with industry experts at Bayer.

Kevin Li



UW/Sanofi HEOR Fellow 2024–2026

PharmD, BA Economics, BS Biochemistry, University of Washington

Kevin recently earned his PharmD from the University of Washington School of Pharmacy. He has built a strong foundation in HEOR through projects in economic modeling, evidence synthesis, and real-world evidence from various educational and professional experiences within the molecular diagnostics, consulting, and pharmaceutical industry fields. Kevin is excited about further advancing his HEOR capabilities and learning from experts at the CHOICE Institute and Sanofi.

Nissen Weisman



UW/Genentech HEOR Fellow 2024–2026

PharmD & BS Pharmacy Studies, University of Connecticut; BA Music, Southern Connecticut State University

Nissen is a recent graduate from the UConn School of Pharmacy. During his time at UConn, Nissen was highly involved with the Health Outcomes, Policy, and Evidence Synthesis (HOPES) Group conducting systematic literature reviews and incorporating economic modeling. Seeing the evidence-generating power that comprehensive literature reviews can accomplish instilled a passion for HEOR. Now he is excited to hone these skills so that he may provide outcomes that will impact patients in meaningful and tangible ways.

Kathryn Perkins



UW/Pfizer HEOR Fellow 2024–2026

PharmD, University of Southern California; BS Molecular Biology, Evergreen State College

Kathryn recently graduated from the University of Southern California with her PharmD and a graduate certificate in Healthcare Analytics and Operations. Her passion for HEOR grew through her experiences in the AMCP P&T Competition, modeling and analytics coursework, and an internship in specialized health outcomes with AbbVie and the AMCP Foundation. Kathryn is motivated by the opportunity for her research to create a direct impact on healthcare decision making and improve patient health outcomes. She is thrilled to develop her skills in data analytics, evidence generation, and HEOR strategy with renowned industry leaders at UW CHOICE and Pfizer.

Olivia Yip



UW/AbbVie HEOR Fellow 2024–2026

PharmD, University of California, San Diego; BS Biochemistry, University of California, Los Angeles

Olivia received her PharmD from UC San Diego, where she developed an interest in HEOR through her involvement in the AMCP P&T competition and various research experiences. These experiences taught her how pharmacoeconomic methods can provide valuable insights into the challenges faced by decision-makers with limited data and assist stakeholders in comparing competing healthcare strategies. She is excited to learn from the distinguished experts at CHOICE and AbbVie, eager to expand her HEOR skill set, and passionate about improving patient access to treatment.

CURRENT STUDENTS

PHD STUDENTS



Shiven Bhardwaj, PharmD, MAS

PhD Student (Year 3)

DISSERTATION TOPIC: TBD

ADVISOR: Doug Barthold



Emma Cousin, PharmD

PhD Student (Year 3)

DISSERTATION TOPIC: TBD

ADVISOR: Sean Sullivan

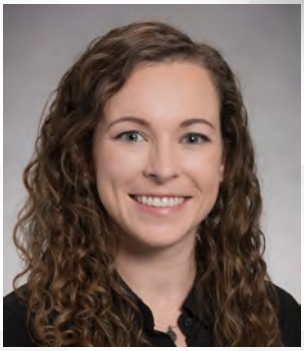


Emily Callen, MPH

PhD Student (Year 3)

DISSERTATION TOPIC: TBD

ADVISOR: Dave Veenstra



Tori Dayer, PharmD

PhD Student (Year 4)

DISSERTATION TOPIC: *Economic Factors in Biosimilar Market Sustainability*

ADVISOR: Sean Sullivan



Hui-Hsuan Chan, MHS

PhD Student (Year 2)

DISSERTATION TOPIC: TBD

ADVISOR: Josh Carlson



Manpreet Dhankhar, MS

PhD Student (Year 2)

DISSERTATION TOPIC: TBD

ADVISOR: Anirban Basu



Zilin Cheng, MS

PhD Student (Year 2)

DISSERTATION TOPIC: TBD

ADVISOR: Jing Li



Ronald Dickerson, MPH, MA

PhD Student (Year 4)

DISSERTATION TOPIC: *Estimating the Impact of Drug Prices and Use on Gun Violence, Economic Costs, and Disease Burden in the United States: An Econometric Approach*

ADVISOR: Dave Veenstra



Zizi Elsisi, MS, BPharm

PhD Student (Year 5)

DISSERTATION TOPIC: *Evaluating the Medicaid subscription-based payment model in Hepatitis C*

ADVISOR: Anirban Basu



Ifechukwu (Ben) Nwogu, MPH

PhD Student (Year 4)

DISSERTATION TOPIC: *Evaluating Newer Therapies for Treatment of Advanced Prostate Cancer*

ADVISOR: Josh Carlson



Julia Fox, MA

PhD Student (Year 3)

DISSERTATION TOPIC: *Cost-Effectiveness Modeling of Opioid Use Disorder in Washington State*

ADVISOR: Kyu Lee



Carlos Pineda-Antunez, MS

PhD Student (Year 2)

DISSERTATION TOPIC: TBD

ADVISOR: Doug Barthold



Karris Jeon, MS

PhD Student (Year 3)

DISSERTATION TOPIC: TBD

ADVISOR: Josh Carlson



Ashley Tabah, MS, MPH

PhD Student (Year 5)

DISSERTATION TOPIC: *Closing the coverage gap: Understanding the spending and adherence effects for cognitively impaired older adults*

ADVISOR: Ryan Hansen



Felipe Montaño-Campos, MA, MS

PhD Student (Year 4)

DISSERTATION TOPIC: *Private Choices in Public Health: A Framework for Economic Epidemiological Modeling of Infectious Diseases*

ADVISOR: Anirban Basu

INSTITUTE UPDATES

THE CHOICE INSTITUTE WELCOMES NOÉMI KREIF AND LAURA HART TO THE FACULTY



Dr. Noémi Kreif, MA, MS, PhD joined the CHOICE Institute and the Department of Pharmacy in March as an Assistant Professor. She joins us most recently from the University of York Centre for Health Economics (CHE) where she was a Senior Research Fellow in Health Economics. Dr. Kreif earned a PhD in Health Economics in 2013 at the London School of Hygiene and Tropical Medicine (LSHTM), University of London (UK). Dr. Kreif also holds an MA in Economics from Central European University, Hungary and an MSc in Economics from Corvinus University of Budapest, Hungary.

Dr. Kreif's current research focuses on combining causal inference and machine learning methods to improve health policy evaluations and economic evaluations by using healthcare data in flexible ways to inform personalized decision making that considers resource and equity constraints. Her most recently awarded research grant allowed her to lead a team that brought together expertise from health economics, causal inference, decision science, and computer science. She is passionate about training a new generation of graduate students and researchers in modern causal inference and computational statistics methods. Dr. Kreif is open to advising MS and Ph.D. students conducting research on a wide range of topics, including using real-world data for comparative effectiveness research and health policy analysis and incorporating machine learning into health outcomes research for stratified decision making.



Dr. Laura Hart, PharmD, MS, BCPS, BCGP joined the Department of Pharmacy as an Assistant Professor in January. She serves as the Assistant Director for Student Engagement in the Plein Center for Geriatric Pharmacy Research, Education and Outreach, and as the David and Anita Bailey Faculty Fellow in Geriatric Pharmacy. Dr. Hart earned her BS and PharmD degrees from the University of Washington, followed by completion of pharmacy practice residencies at the Veterans Affairs Puget Sound Health Care System and the UW School of Pharmacy (UWSOP). During her residency with UWSOP she earned her MS degree through the CHOICE Institute. Most recently, Dr. Hart served as an Assistant Clinical Professor at the University of California (UC), San Diego Skaggs School of Pharmacy and Pharmaceutical Sciences (SSPPS).

Dr. Hart's research interests are broadly related to optimizing medication use in older adults, including the safe use of over-the-counter medications, deprescribing, and falls prevention. She is also passionate about efforts related to diversity, equity, and inclusion, with a particular focus on the LGBTQIA+ community. Dr. Hart currently serves as co-chair of the ASCP Medication Safety and Transitions of Care Workgroup. She has been honored for her contributions to the profession of pharmacy, including receiving the ASCP Next Gen-Rx Award, UWSOP Distinguished Alumni Award: Early Career Achievement Award, Rho Chi Alumni Award, and UC San Diego SSPPS Faculty of the Year Award. Dr. Hart is excited to return to UW and looks forward to collaborating with CHOICE faculty members and trainees on geriatric pharmacotherapy outcomes research.

SCHOLARGPS WORLDWIDE RANKING: UW SOP AMONG TOP 3 HIGHEST-RANKED INSTITUTES

Dr. Sean Sullivan:
Among Top 7 high-ranked scholar in Managed Care

Dr. Lou Garrison:
Among Top 4 high-ranked scholar in Health Economics

Dr. Anirban Basu:
Among Top 6 high-ranked scholar in Health Economics

< LEARN MORE

SAVE THE DATE WPEC 2025

THE CHOICE INSTITUTE AT THE UNIVERSITY OF WASHINGTON IS PROUD TO BE HOSTING THE 2025 WESTERN PHARMACOECONOMICS CONFERENCE (WPEC)

MARCH 31 – APRIL 2, 2025

UNIVERSITY OF WASHINGTON CAMPUS

SEATTLE, WA

Visit our website to learn more about sponsorship opportunities and to stay in the loop with any conference updates.
https://sop.washington.edu/choice/research/seminars-events/wpec2025/



CONTACT OUR EXECUTIVE PLANNING TEAM:

Anirban Basu, MS, PhD
Professor & Stergachis Family Endowed Director
basua@uw.edu

Josh Carlson, MPH, PhD
Professor & Graduate Program Director
carlsojj@uw.edu

WPEC 2025 SCIENTIFIC COMMITTEE:

David Veenstra (co-chair)
Professor

Douglas Barthold (co-chair)
Research Associate Professor

Shiven Bhardwaj
HEOR PhD Student

Emily Callen
HEOR PhD Student

Yunjoo Karris Jeon
HEOR PhD Student

2023 ANNUAL SYMPOSIUM RECAP: HEALTH EQUITY IN HEOR

The 5th Annual Symposium at The CHOICE Institute, held on October 12, 2023, centered around "Artificial Intelligence, Real Consequences: The Side Effects of AI in Healthcare." Moderated by Aasthaa Bansal, PhD, the event featured insightful discussions on the potential benefits and unintended consequences of AI technologies in medical practice and health economics and outcomes research through presentations by three experts.

Michael Willingham, Director of Regulatory Affairs at Google Health, delivered a talk titled "Considerations for Large Language Models and Generative AI in Medical Applications." He explored the transformative potential of large language models (LLMs) like GPT-4 and Bard in clinical documentation, research summarization, and clinical decision-making chatbots. Willingham highlighted the unique risks these models pose in patient care, since these models are trained differently from previous AI and machine learning models. He presented case studies from Google's research on Med-PaLM, a LLM tailored for medical applications, illustrating both the promise and challenges of deploying advanced AI in healthcare.

Noémi Kreif, PhD, Assistant Professor at The CHOICE Institute, presented "Machine Learning to Answer Causal Questions: How to Avoid Misleading Interpretation?" Dr. Kreif discussed the pitfalls of using machine learning prediction models for treatment decisions, citing examples from clinical medicine. She emphasized the importance of causal machine learning in health policy research, demonstrating through case studies, including evaluations of health insurance and cash transfer initiatives in Indonesia, how careful adjustment for confounding factors can mitigate misleading recommendations.

Judy Gichoya, MD, MS, Associate Professor at Emory University School of Medicine, addressed "Hidden in Plain Sight: Harnessing AI's Ability for Population Health from Medical Images." Dr. Gichoya explored AI's potential in deriving population health insights from medical images, stressing the need to address biases for equitable outcomes across diverse populations. She discussed applications in radiology for screening and diagnosing diseases like diabetes. Dr. Gichoya concluded with a call for enhanced AI methodologies and interdisciplinary collaborations to advance population health management.

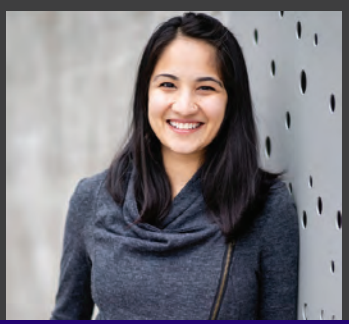
Key takeaways included the critical need for ethical AI deployment in healthcare, the potential benefits of advanced AI models like LLMs tempered by unique risks, and the role of rigorous research methodologies in ensuring AI-driven decisions are reliable and equitable. The symposium underscored AI's evolving role in transforming healthcare while emphasizing the importance of responsible implementation and continuous ethical oversight.











2024 **ANNUAL RETREAT**

Written by Marina Gano, Julia Fox, Emma Cousin, Manpreet Dhankhar, and Jing Li

We were thrilled to host the whole Institute for our 2024 Annual Retreat with the theme "DCE: Diverse CHOICE Experience" at the UW Waterfront Activities Center. This year's planning team consisted of Marina Gano, Julia Fox, Emma Cousin, Manpreet Dhankhar, and Jing Li. We spent the day taking part in various activities with the goal of promoting and celebrating diversity within CHOICE. Some of these are proud traditions of past years' retreat (e.g. speed networking, CHOICE Trivia), and some are new activities specifically designed for this year's theme (i.e. Identity Wheel & Career Maps, Free Roaming Activities that included group arts and pinning cities of residence on a world map). These activities led to both fun and reflection among participants.

Additionally, Anirban Basu, the CHOICE Institute Director spoke about the state of the CHOICE Institute. Josh Carlson, Director of the Graduate Program shared the graduate program updates. Ben Nwogu, the ISPOR Student Chapter President shared the ISPOR Student Chapter updates. As representatives of the WPEC 2025 Scientific Committee, Douglas Barthold and Emily Callen led a brainstorming activity on the possible scientific themes of WPEC 2025.

Lastly, Jay Panyam, our new School of Pharmacy Dean joined us for the Awards & Recognitions Celebration and the social reception. The social reception was hosted at the same location on the private patio outside of the Waterfront Activities Center room overlooking the beautiful Union Bay. We were honored to have CHOICE family, friends, partners, and children join us to close out a day of celebrating and enhancing the diversity experience of our CHOICE community.











PICTURED RIGHT: CHOICE STUDENTS AND FACULTY DURING THE ANNUAL RETREAT. PHOTOS BY OLIVIA CARSON.

2024 **ISPOR ANNUAL CONFERENCE**

Always a highlight of our year — this year's ISPOR Annual Conference was in sunny Atlanta, Georgia. We extend a sincere thank you to all those who joined us on Monday night at Wild Leap Atlanta for our annual alumni & friends reception. At this event, we connected with nearly 150 alumni, academic and industry colleagues, members of our Corporate Advisory Board and more.

Our students, faculty, and alumni made a great showing at the conference, through leading sessions and presenting their work as posters. On this page are a few key highlights and photos.

Of note, Professor & Stergachis Family Endowed Director, Anirban Basu, spoke at the final plenary session on Wednesday, titled, AI Enabling Whole Health: Opportunities and Challenges for HEOR and HTA. Key takeaways from this session can be found on a piece by The Evidence Base, linked here: **https://www.evidencebaseonline.com/ispor-2024-inside-the-third-plenary-leveraging-ai-in-heor-and-hta-to-enable-whole-health/**

Looking forward to seeing many of you again at ISPOR EU 2024 in Barcelona and at the ISPOR 2025 in Montreal!

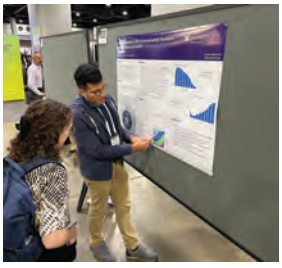








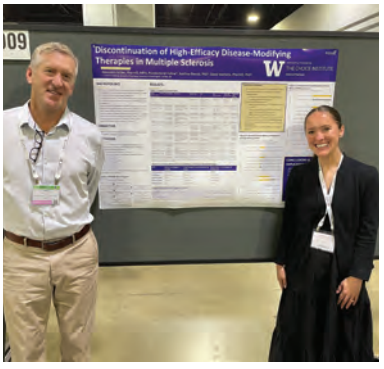














ABOVE: ALUMNI, FACULTY, AND STUDENT ACTIVITIES AT ISPOR ATLANTA 2024

ISPOR CHAPTER UPDATE

The University of Washington ISPOR Student Chapter had an amazingly productive 2023/2024 school year. The Chapter Board, consisting of Ben Nwogu (President), Manpreet Dhankhar (Vice President), Jennifer Deem (Secretary), and Felipe Montano-Campos (Treasurer), successfully planned and executed several events. These events included social activities, career workshops, and networking sessions, all under the guidance of Dr. David Veenstra, the faculty advisor.

In the Fall quarter, the Chapter kicked off he year with a Welcome Social Reception at Ravenna Brewing, where students and friends shared their summer experiences and expectations for the new school year, among other fun conversations. Following this, we held our Annual UW ISPOR Seminar and hosted our second-ever board elections. In a collaborative spirit, the Chapter jointly organized a Halloween-themed Happy Hour with the UW AMCP (Academy of Managed Care Pharmacy) at Big Time Brewery.

In the Winter quarter, we hosted an engaging industry career talk titled "Market Access Strategies: Understanding the Value of Care in the Pharmaceutical Industry" The keynote speaker, Edouard Lassalle, Associate Director of Market Access Operations & Analytics at Jazz Pharmaceuticals, provided invaluable insights into the complex process of launching pharmaceutical products in the US and international markets. His presentation covered country-specific differences in value assessment, pricing strategies, and reimbursement methods.

In the Spring quarter, the Chapter was well-represented at the 2024 ISPOR conference in Atlanta, GA, with many students delivering poster presentations. Locally, the Chapter organized an arcade gaming event at the Ice Box Arcade, which was an evening of fun, games, and excitement. We also hosted a career workshop titled "Mastering the Pharma Job Market" with Nina Hill, a CHOICE alumna and Senior Director of Health Economics and Outcomes Research (HEOR) at Puma Biotechnology. She shared powerful insights into the pharma job market, from finding the right job to preparing for interviews and salary negotiation. Finally, the Chapter hosted an interactive networking session with Rob Abbott, the CEO of ISPOR, providing a fantastic opportunity for our students to engage with the global ISPOR leadership and strengthen our Chapter's affiliation.

For the 2024/2025 academic year, Manpreet Dhankhar will succeed Ben Nwogu as President of the Chapter. Additionally, Noemi Kreif will take on the role of faculty advisor for the UW ISPOR Chapter. We eagerly anticipate the upcoming school year.





ONLINE CERTIFICATE PROGRAMS UPDATE

The CHOICE Institute is excited to now offer **two** certificates that advance skills in health economics and outcomes research. Offered in partnership with UW Professional & Continuing Education, our certificates are designed for working professionals and cover a wide range of topics in HEOR. Courses are online and asynchronous, allowing for maximum flexibility.

CERTIFICATE IN DATA SCIENCE IN HEALTH ECONOMICS & OUTCOMES RESEARCH (NEW)

We're thrilled to launch the Certificate in Data Science in Health Economics and Outcomes Research. This new program is designed for professionals who want to understand how big data is used to make decisions in health and policy settings, including payer organizations, health industries, and government agencies, as well as life sciences, pharmaceutical and biomedical industries.

This accessible three-course certificate provides comprehensive exposure to data science concepts in the context of HEOR. A unique feature is its focus on data science concepts and their application in healthcare data analysis rather than coding or complex statistical derivations. Lessons cover fundamentals like working with real-world data, causal inference methods, and machine learning principles and how these principles and methods are applied in the HEOR field.

"This journey has been incredibly enriching, providing me with deep insights into the economic and clinical outcomes of healthcare interventions. I am grateful for the opportunity to learn from esteemed faculty members and to collaborate with a cohort of talented professionals. This program has equipped me with the skills to drive evidence-based decision-making and improve healthcare delivery."

- **Anas Alonezan, MD**, 2024 GRADUATE OF THE CERTIFICATE IN HEALTH ECONOMICS & OUTCOMES RESEARCH

THIS 9-MONTH PROGRAM OFFERS ONE COURSE PER ACADEMIC QUARTER:

FALL QUARTER:
Principles of Data Science & Health Care Data, taught by Dr. Beth Devine

WINTER QUARTER:
Comparative Effectiveness Analysis with Routine Care Data, taught by Dr. Anirban Basu

SPRING QUARTER:
Principles of Machine Learning & Applications in Health Economics & Outcomes Research, taught by Dr. Aasthaa Bansal

Courses include recorded lectures, readings, discussions, individual capstone projects, and opportunities to interact virtually with faculty in optional group meetings each quarter. Students will take away knowledge to evaluate data science projects and make evidence-based decisions that advance health research, policy, and outcomes.

CERTIFICATE IN HEALTH ECONOMICS & OUTCOMES RESEARCH

The new certificate complements our well-established Certificate in Health Economics and Outcomes Research. Entering its 13th year, the Certificate in HEOR continues to be a highly regarded program for working professionals looking to learn more about HEOR. The online program has trained over 550 students from a wide range of health care settings including payer organizations, health insurance companies, government, pharmaceutical and biomedical industries.

Students learn the key economic concepts and analytical tools of human economic behavior, explore the current state of the art in the economic evaluation of health care technologies, and are introduced to the principles and methods of HTA practice in the U.S. and around the world.

Students have enrolled from all around the U.S. and from over 30 countries. Courses are continually updated to incorporate changing technologies, economies, and policy. Student evaluations have remained positive over the years. After completing the certificate, students can stay connected through the LinkedIn alumni group. Faculty enjoy getting to meet certificate students at the CHOICE alumni reception during the annual ISPOR conference.

Applications

We are accepting applications to both programs for the 2024–2025 academic year. The deadline to apply is September 11, 2024. Courses start September 25, 2024.

Certificate in Data Science in Health Economics and Outcomes Research:

https://www.pce.uw.edu/certifi ates/data-science-health-economics-outcomes-research

Certificate in Health Economics and Outcomes Research:
https://www.pce.uw.edu/certificates/health-economics-and-outcomes-research

Questions? Contact Leana de la Torre, Certificate Program Manager, at torre@uw.edu or UWPCE at enroll@pce.uw.edu

THIS 9-MONTH PROGRAM OFFERS ONE COURSE PER ACADEMIC QUARTER:

FALL QUARTER:
Principles of Health Economics, taught by Dr. Lou Garrison and Dr. Douglas Barthold

WINTER QUARTER:
Economic Evaluation in Health and Medicine, taught by Dr. David Veenstra and Dr. Josh Carlson

SPRING QUARTER:
Practice of Health Technology Assessment in a Global Environment, taught by Dr. Sean Sullivan and Dr. Beth Devine

Each quarter there are optional group meetings where students can interact virtually with faculty. Students work in groups throughout the program on a capstone project, which applies their learning to an assigned country. Students will take away knowledge to address problems of resource allocation and improve decision-making in complex health settings.

THE PROGRAM IN HEALTH ECONOMICS AND OUTCOMES METHODOLOGY (PHEnOM)

The Program in Health Economics and Outcomes Research Methodology (PHEnOM) is a seminar series aiming to facilitate intellectual activities in the fields of health economics, econometrics, and methodology development in outcomes research and to engage researchers in allied fields. Key competencies of PHEnOM affiliates include health econometrics, health policy evaluation, applications in health capital, aging and disability, organization of healthcare, simulation methods, and comparative and cost-effectiveness analysis. It is co-directed by Dr. Douglas Barthold, Research Associate Professor and Dr. Jing Li, Associate Professor at CHOICE.

PHEnOM hosts a weekly seminar on Wednesdays, from 3–4:30PM, in the Department of Pharmacy Conference room (H371), with virtual attendance also available. The seminar series has attracted a growing audience from both within and outside of UW. In Winter 2024, we organized the first student- and early career-focused seminar series that featured early career and doctoral student presenters from different departments/schools across the UW campus (e.g. CHOICE, Public Health, Economics and Business) and institutions across the country (Harvard, U Texas-Austin and University of North Carolina). It was a successful endeavor to expose our students to a wide range of high-quality research conducted by their peers and for the presenters to get valuable feedback.

Some highlights of the last year were presentations from Graeme Peterson (Harvard) on the geography of health disparities in Medicare, and from Dr. Lauren Nicholas (U Colorado) on experimentally measuring how surrogates make treatment decisions for patients with dementia. We are looking forward to more interesting and impactful research at the PHEnOM Seminar during the 2024–2025 academic year, and especially welcome nomination and self-nomination of speakers internal and external to UW, including those at the early stage of their career or in training.

Please contact Dr. Barthold (barthold@uw.edu) or Dr. Li (jli0321@uw.edu) if you are interested in presenting at PHEnOM or subscribing to the PHEnOM-info email list.







THE CHOICE INSTITUTE PARTNERS WITH TRUVETA TO ADVANCE CRITICAL HEALTH ECONOMIC AND OUTCOMES RESEARCH

TRUVETA WILL ENABLE THE CHOICE INSTITUTE WITH COMPLETE, TIMELY, AND CLEAN REAL-WORLD DATA TO IMPROVE PATIENT CARE AND ADVANCE MEDICINE

The Comparative Health Outcomes, Policy, and Economics (CHOICE) Institute at the University of Washington's School of Pharmacy has partnered with Truveta to use its electronic health record (EHR) data to drive comparative, cost-effectiveness, and outcomes research.

Researchers at The CHOICE Institute will use Truveta data to gain access to real-world EHR data from a growing collective of 30 health systems across the country. The CHOICE team will use these data to generate insights across various therapeutic areas, including diabetes, dementia, sickle cell disease, pain management, cystic fibrosis, hepatitis C, and others. Researchers will study the intersection between health policy, chronic disease management, and cognition; the value of care for conditions influencing healthy aging; vaccinations; antibiotic use; location of care; patient reimbursement; health econometrics and comparative effectiveness; and more.

Truveta offers the most complete, timely, and clean EHR data from more than 100 million patients across 30 US health systems, empowering researchers to study all diseases, drugs, and devices. Truveta Data is representative of inpatient and outpatient care from over 900 hospitals and 20,000 clinics. Truveta Data is updated daily for the most current view of patient care. By providing a complete, timely view of the patient journey, including clinical notes and medical images, Truveta enables researchers to study comparative and cost effectiveness of therapies and outcomes for representative populations. Truveta Studio allows researchers to explore research questions and create new studies and conduct subsequent internally or grant-funded research. Researchers can quickly iterate study design and feasibility, with the ability to modify inclusion and exclusion criteria in real time for representative and precise populations, including underrepresented communities and those with rare diseases.

"Truveta offers a curated, regulatory-grade EHR database that covers almost a third of care delivered in the United States. We are excited to partner with them and position our team to conduct a wide range of advanced research across health economics, outcomes, and comparative effectiveness," said Anirban Basu, PhD, Stergachis Family Endowed Director and Professor of Health Economics at The CHOICE Institute.

"We are honored to partner with an innovative academic research organization like The CHOICE Institute, working toward a shared mission of saving lives with data," said Truveta Chief Medical Officer and Co-founder Ryan Ahern, MD, MPH. "Through the power of real-world data, we can unlock critical patient insights to researchers, enabling important health economic and outcomes research to advance medicine and improve patient care."



PROFESSOR EMERITUS GARRISON SPEAKS AT HTA CONFERENCE IN KAZAKHSTAN





Professor Emeritus Lou Garrison recently gave three talks at a health technology assessment conference sponsored by KazSPOR and HTAi in Astana, Kazakhstan. He gave two short talks and a one-hour lecture on the basic economics of innovative pharmaceuticals. He began working three years ago with Dr. Alex Kostyuk, the founder and leader of HTA there, and with Dr. Alima Almadiyeva, who he mentored during her Fulbright-Humphreys fellowship at Emery University. His talks dovetail with his efforts as chair of a new ISPOR Special Interest Group on Global Access to Medical Innovation. He says that: "We urgently need to address the access or drug lag to innovative medicines in low- and middle-income countries by encouraging efficient HTA and establishing more a more effective approach to global differential pricing." This SIG is new and had a recruitment/information session at the ISPOR meeting in Atlanta. Lou was also able to visit the "Little Switzerland" by Lake Buraby, north of Astana, where he encountered a group of students eager to meet a UW faculty member.

CHOICE FC INAUGURAL SEASON

The CHOICE Institute founded a soccer team, "CHOICE FC," in the UW Faculty/Staff/Graduate Students Intramural League. The team finished in 3rd place in the league, with a final record of 6 wins and 2 losses. There were spectacular goals, and the team had memorable victories over many other academic units from around UW, including the sociologists, fishery scientists, chemists, and the law school. A highlight of the season was Research Associate Professor Doug Barthold, passing to Carlos who then scored a goal with a picture-perfect bicycle kick. The team's members include Doug Barthold, Felipe Montano-Campos, Emily Callen, Carlos Pineda Antunez, Filmon Haile, and Jake Earl. The team looks forward to their second season!



CHOICE FACULTY CO-AUTHOR STUDY ON HOW **MEDICARE IS OVERPAYING FOR GENERIC DRUGS**

Disturbing financial practice by Medicare Part D sponsors revealed in University of Washington study of data from the Centers for Medicaid and Medicare Services

Medicare is the single largest provider of health insurance in the United States, serving 63.8 million senior citizens as of 2022. Three-quarters of these recipients are enrolled in optional Medicare Part D plans, which provide outpatient prescription drug coverage to seniors through private insurance companies. In 2022, Medicare paid more than $160 Billion for prescription drugs, making it the single largest payer of pharmaceuticals in the US. While Medicare is meant to keep healthcare affordable for seniors, millions of Americans still face steep costs for prescription medications. Researchers from the CHOICE Institute, University of Washington, the University of California San Diego and West Health Policy Center have now explained part of the problem.

The researchers found evidence that the private insurers that sponsor Medicare Part D (through their Pharmacy Benefit Managers [PBMs]) are inflating the costs of some generic drugs by overpaying pharmacies. The findings are published in the December 5, 2023 issue of the Journal of the American Medical Association. The researchers gathered and analyzed data from the Centers for Medicare and Medicaid Services (CMS), focusing on spending and reimbursement data for the 50 generic drugs that Medicare Part D spent the most on in 2021. They found that some Medicare Part D sponsors were reimbursing at much higher rates than what pharmacies spent to acquire the drugs.

To fill prescriptions, pharmacies purchase drugs wholesale and are reimbursed by insurers. In order to be reimbursed, pharmacies must adhere to contractual obligations that allow insurers to take back money in certain circumstances, such as when the reimbursement to the pharmacy is higher than a certain threshold, or if the pharmacy's cost of acquisition is very low. These 'clawbacks' by Part D Plans can be financially devastating to a pharmacy.

"It doesn't make sense that insurers would overpay for drugs, then use claw backs to retroactively adjust payments after the patient has paid their co-payment," said co-author Sean D. Sullivan, PhD, professor of pharmacy at the University of Washington. "This practice is opaque and ultimately harms patients and pharmacies."





That harm also extends to relationships between pharmacists and their patients, according to UW Department of Pharmacy Professor and Interim Chair Ryan N. Hansen, PharmD, PhD.

"Not only have PBM business practices bankrupted community pharmacies, they damage the important relationships between pharmacists and their patients by creating unaffordable out-of-pocket costs, resulting in difficult decisions to forego important treatments at the pharmacy counter," said Hansen.

Generic over-reimbursement is one of the areas targeted for reform by the United States Senate, after anecdotal evidence submitted earlier this year by the non-profit manufacturer CivicaRx raised concerns about Part D sponsors potentially over-reimbursing pharmacies for abiraterone, a cancer drug. The concern: that patients are ultimately impacted by these reimbursement practices because of how out-of-pocket costs (e.g., co-payments and deductibles) are calculated.

"For instance, if a patient pays 30% as a co-payment for a drug, that 30% would be applied by the PBM to the inflated price, not the price the pharmacy paid or the price the PBM pays the pharmacy, which means higher out-of-pocket costs for seniors," said corresponding author Inmaculada Hernandez, PharmD, PhD, professor at UC San Diego Skaggs School of Pharmacy and Pharmaceutical Science. "This is the first study to investigate whether these PBM practices actually take place in the Medicare Part D program," added Hernandez.

(continued on next page)

CHOICE FACULTY CO-AUTHOR STUDY ON HOW **MEDICARE IS OVERPAYING FOR GENERIC DRUGS** continued

"The results are alarming," said Hernandez. "We are talking about significant markups in some cases."

In one of the most dramatic examples, the researchers found that insurers were reimbursing pharmacies an average of $126 per tablet for a cancer drug that cost $4.20 per tablet to the pharmacy. This corresponds to an average markup of 3,000%, or $3,600 per 30-day prescription. Some insurers paid even more.

"Seniors are clearly paying more for their medications as a result of these markups," said Hernandez. "More research is needed to confirm the scope of these practices, but the evidence we present in the paper is concerning."

Full link to study: **https://jamanetwork.com/journals/jama/newonline**

Co-authors include: Nico Gabriel at UC San Diego, Anna Kaltenboeck at ATI Advisory, and Cristina Boccuti at West Health Policy Center.

This study was funded by West Health Policy Center.







CHOICE RESEARCH ON THE **MEDICARE DRUG PRICE NEGOTIATION PROGRAM**

In December, PhD student Emma Cousin, PharmD, Professor Sean D. Sullivan, BScPharm, PhD, and their collaborators received one of the three 2023 PhRMA Foundation Challenge Awards for their manuscript titled "Medicare Drug Price Negotiation: The Complexities of Selecting Therapeutic Alternatives for Estimating Comparative Effectiveness". This research identifies clinical comparators to the first 10 drugs selected for the Medicare Drug Price Negotiation Program. The authors also describe the challenges that the Center for Medicaid and Medicare Services (CMS) may face in selecting therapeutic alternatives, and its implications in estimating comparative effectiveness to determine the maximum fair price (MFP).
As a follow-up to this work, Drs. Cousin and Sullivan recently published another manuscript titled "Price Benchmarks of Drugs Selected for Medicare Price Negotiation and Their Therapeutic Alternatives". In this paper, they estimate key price benchmarks that are involved in deriving the initial price offer for the first ten drugs selected for the Medicare Drug Price Negotiation Program. The result from this work demonstrates that the savings achieved from Medicare negotiation program varies significantly across drug products. Both manuscripts were done in collaboration with the University of Washington, the University of California, San Diego, The London School of Economics & Political Science, and funded by The Commonwealth Fund.

In addition to this work, Drs. Cousin and Sullivan, alongside Dr. Inmaculada Hernandez from UC San Diego, recently presented an ISPOR Webinar in July titled "IRA Part II: Next 15 For Medicare Drug Price Negotiations in 2025". During the webinar, the authors predicted the next set of Part D drugs eligible for the Medicare negotiation program based on projected 2023 spending and discussed why other drugs with comparable gross spending are considered ineligible. This webinar was attended by 260 participants and is available as a recording via the Learning Lab on the ISPOR website.

PARTNERING WITH COMMUNITY PHARMACIES TO ENHANCE ACCESS TO LONG-ACTING INJECTABLE ANTIPSYCHOTICS IN WASHINGTON STATE

Assistant Professor Clayton English and his co-investigators, Associate Professor Jenny Bacci and Professor Andy Stergachis, received a 2023 Garvey Institute for Brain Health Solutions (GIBHS) Innovation Grant award. Their project is titled, "Partnering with community pharmacies to enhance access to long-acting injectable antipsychotics (LAIAs) in Washington State."

Medication nonadherence is common among patients with serious mental illness, including schizophrenia. The use of long-acting injectable antipsychotics (LAIAs) for schizophrenia is an evidence-based practice that improves medication adherence, decreases symptomatic recurrence and reduces hospitalizations. However, patients and clinicians often face several challenges in access and coordination resulting in the underutilization of LAIAs in care.

Administering LAIAs at community pharmacies could potentially increase accessibility, reduce barriers for treatment and improve patient outcomes. This project aims to assess the fit or compatibility of LAIA administration in community pharmacies. We are conducting a survey of community pharmacy staff and behavioral healthcare providers in Washington State to assess the acceptability, appropriateness and feasibility of LAIA administration in community pharmacies. If LAIA administration at community pharmacies is found to be a good fit, the next steps will be to develop strategies to support implementation. A scalable and adoptable model for administering LAIAs at community pharmacies could have substantial impacts on public health through increasing access to treatment and expanding behavioral health services at the community level and in rural areas.

The UW Medicine Garvey Institute for Brain Health Solutions is fundeding 12 new partnerships that will develop and test innovative ways to extend the impact of mental health professionals in Washington. The focus on partnerships was driven by a shortage of mental health care specialists in Washington. More than half of Washington's 39 counties lack a single practicing psychiatrist or psychologist, leaving many people with untreated mental health conditions. Local philanthropists Lynn and Mike Garvey founded the Garvey Institute in 2019 with a $50 million donation to UW Medicine with a goal to fast-track treatments for patients with mental health, addiction and other brain health problems.

Source: UW Medicine, Garvey Institute for Brain Health Solutions, Department News, November 30, 2023.







UW/ICER COLLABORATION UPDATE **2024**

Written by Josh Carlson

In 2016, the UW CHOICE Institute and the Institute for Clinical and Economic Review (ICER) formed a partnership establishing the first multi review academic site. This relationship has continued for 8 years and 4 additional contracts. The current team is comprised of Josh Carlson (PI), Linda Luu (research scientist/lead modeler), Kangho Suh (lead modeler), an Assistant Professor at the University of Pittsburg and a CHOICE alumnus, and HEOR PhD Students Ronald Dickerson (Research Assistant) and Hui-Hsuan Chan. Our current contract runs through 12/31/25.

To date we have completed or are working on economic evaluations on 25 topics: Special Assessment to Inform CMS Drug Price Negotiation: COPD (ongoing), Anemia in Myelodysplastic Syndrome (2024), Paroxysmal Nocturnal Hemoglobinuria (2024), Metachromatic Leukodystrophy (2023, Amyotrophic Lateral Sclerosis (2022), Chemotherapy induced myelosuppression (2022,)Anemia in chronic kidney disease (2020–2021), Supervised Injection sites (2020), Ulcerative Colitis (2020), nonalcoholic steatohepatitis (2020 and 2023), Type II diabetes (2019, 2021), peanut allergy (2019), multiple myeloma (2016), non-small cell lung cancer (2016), plaque psoriasis (2016 and 2018), relapsing remitting multiple sclerosis (2017), secondary progressive multiple sclerosis (2019), atopic dermatitis (2017, 2021), osteoporosis (2017), RPE65-mediated inherited retinal disease (2018), Hemophilia A (2018), prostate cancer (2018), and hereditary angioedema (2018, 2021). Our work has been disseminated at ICER public meetings, as part of ICER's full topic reports, in conference proceedings (9 abstracts), and in peer reviewed journals (18 published).

In recent evaluations, we performed a review of treatments for Metachromatic Leukodystrophy (MLD). For MLD, a virtual public meeting was held for the California Technology Assessment Forum (CTAF) on September 29, 2023. We found that over a lifetime time horizon, treatment with atidarsagene autotemcel (Lenmeldy, "arsa-cel", Orchard Therapeutics) resulted in incremental quality adjusted life years (QALYs) and equal value life years (evLYs) of approximately 21.45 and 18.22, respectively. The health benefit price benchmark (HBPB) for Lenmeldy ranged from $2.2 to $3.9 million.



We also recently completed a review for Paroxysmal Nocturnal Hemoglobinuria (PNH). A virtual public meeting was held for the CTAF on February 16, 2024. We found that over a lifetime time horizon, treatment with iptacopan (Fabhalta ®, Novartis) resulted in incremental QALYs and evLYs of approximately 0.15. Treatment with Danicopan + Ravulizumab resulted in incremental QALYs and evLYs of approximately 0.06. The health benefit price benchmark (HBPB) for iptacopan was $178,000 to $180,000 annually and for Add-on Danicopan it was $12,300 to $13,100 annually.

The UW team continues to be part of the ICER interactive modeler initiative that allows decision-makers within payers, pharmaceutical companies, and other stakeholders to access cost-effectiveness models via a web enabled platform. There are now 16 UW models available through the IM platform.

In summary, our collaboration with ICER continues, providing valuable research opportunities for faculty and students and contributing to timely and impactful conversations about the value of emerging products in the U.S. We look forward to continued collaboration with ICER and the many companies developing innovative medical products.

UTILIZATION OF LOW- AND HIGH-VALUE HEALTHCARE BY INDIVIDUALS WITH AND WITHOUT COGNITIVE IMPAIRMENT

Cognitive impairment and dementia have rising prevalence and impact both healthcare utilization and 'the lives of older adults and their families. Receipt of low-value care and under-utilization of high-value care by people living with these cognitive disorders may contribute to negative consequences on patient health, health system efficiency, and societal welfare.

In a study recently published in American Journal of Managed Care, Dr. Douglas Barthold and collaborators from The CHOICE Institute examined if the onset of cognitive decline is associated with the use of specific high and low value services, among Medicare beneficiaries from 1996-2018. The results show suboptimal delivery of high value services and that low-value care use was common among all older adults (4-13%). Of additional concern was the finding that cognitive decline was associated with decreased probability of receipt of 3 high-value services (glucose-lowering drugs, statins, and antiresorptive therapy).

Said Mark A. Fendrick, MD, Director of the University of Michigan's Center for Value-Based Insurance Design, and the study's senior author: "Given the persistence of these undesirable outcomes, we should more seriously consider a strategy that reallocates the substantial workforce effort and financial resources currently devoted to the delivery of low-value care to enhance access to and affordability of high-value services, especially for vulnerable populations such as those with cognitive impairment."

The relatively common use of low-value services, regardless of cognitive status, suggests persistent waste in the health care of older adults. Lower use of 3 high-value services after cognitive decline is worrisome because these individuals would be expected to benefit from these services. A drop in the use of high-value services may represent a missed opportunity because optimal (evidence-based) care could result in improvement of health, quality of life, and well-being.

Barthold D, Jiang S, Basu A, Phelan EA, Thielke ST, Borson S, Fendrick AM, "Utilization of low- and high-value healthcare by individuals with and without cognitive impairment," American Journal of Managed Care, 2024; 30(7).





EXPLORING THE POTENTIAL BENEFITS OF DELAYING SEASONAL INFLUENZA VACCINE SELECTIONS

In a retrospective modeling study, Assistant Professor Kyu Lee and her co-authors examined the potential advantages of delaying seasonal influenza vaccine selections for the Northern Hemisphere. The study aimed to assess the impact of updating vaccine formulation schedules to improve antigenic similarity between vaccine viruses and circulating viruses, which is crucial for high vaccine effectiveness.

Current vaccine formulation decisions for the Northern Hemisphere are made in February, but advancements in non-egg-based vaccine production technologies could allow for more flexibility. The study identified past influenza seasons where the dominant viruses continued to increase in prevalence after the formulation decisions were made, leading to a mismatch between the vaccine and circulating viruses.

Using a dynamic compartmental model of influenza transmission, the researchers evaluated the potential benefits of updating vaccine decisions with more antigenically similar viruses. They focused on A(H3N2) and B/Victoria in the 2014/15 and 2019/20 influenza season, respectively, where delayed formulation decisions could have improved the match between the vaccine and circulating viruses.

The model demonstrated that updating the A(H3N2) vaccine in the 2014/15 season could have prevented between 5,000 and 65,000 influenza-related hospitalizations in the United States. However, updating the B/Victoria vaccine component in the 2019/20 season did not significantly affect the in fluenza burden.

The study concluded that revising current vaccine selection timelines, facilitated by rapid vaccine production, could provide substantial epidemiological benefits. Specifi ally, delaying vaccine virus selection decisions could improve the antigenic match between the vaccine and circulating viruses, thereby enhancing vaccine effectiveness and reducing the influenza burden.

Lee K, Williams KV, Englund JA, Sullivan SG. The potential benefits of delaying seasonal influenza vaccine selections for the Northern Hemisphere: a retrospective modeling study in the United States. J Infect Dis. 2023 Nov 29:jiad541. doi: 10.1093/infdis/jiad541. Epub ahead of print. PMID: 38019883.

PATIENT OUT-OF-POCKET COSTS FOR TYPE 2 DIABETES MEDICATIONS WHEN AGING INTO MEDICARE

For people with type 2 diabetes, out-of-pocket medication costs influence medication choice, adherence, and overall diabetes management and progression. Little is known about how these costs change as insured people enter Medicare at age 65, when coinsurance in the coverage gap and catastrophic phases of Part D coverage can be driven especially high by use of insulin and newer, branded medications (e.g., DPP4 inhibitors, GLP1 agonists, and SGLT2 inhibitors).

In a new study published in JAMA Network Open, CHOICE Institute researchers examined the effects of entering Medicare on out-of-pocket drug costs for people with type 2 diabetes. Using a regression discontinuity design, they found dramatic increases in out-of-pocket costs upon reaching age 65 and entering Medicare. On average, spending increased from $77 to $109 per quarter, and at the 95th percentile, spending increased from $337 to $508 per quarter. The analyses also showed that drug utilization decreased slightly at age 65 (5% fewer claims per quarter).

The increase in spending upon entering Medicare is likely caused by patient coinsurance in the coverage gap and catastrophic coverages phases of Medicare Part D. The increased patient cost burden at age 65 and a modest reduction in overall diabetes drug utilization suggest that as people with diabetes age into Medicare, there is potentially an increase in non-adherence and diabetes complications.

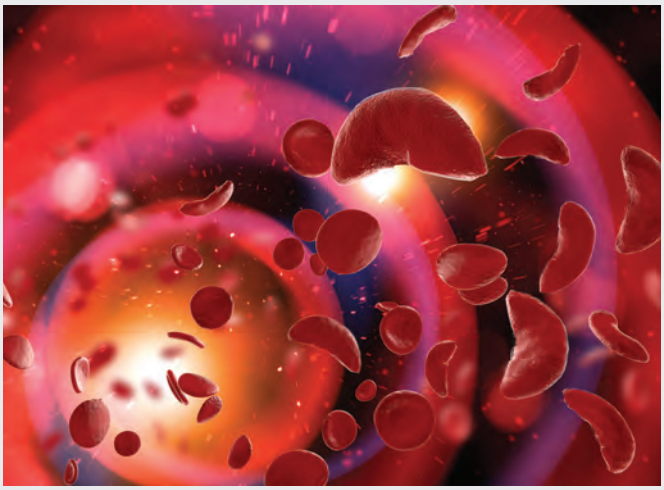
Barthold D, Li J, Basu A, “Patient out-of-pocket costs for type 2 diabetes medications when aging into Medicare,” JAMA Network Open, 2024; 7(7): e2420724. DOI: 10.1001/jamanetworkopen.2024.20724

EVALUATING GENE THERAPIES UNDER THE NIH CURE SICKLE CELL (#CURESC) INITIATIVE

The Cure Sickle Cell Initiative (#CureSC, https://curesickle.org/) is a collaborative, patient-focused research effort designed to accelerate promising genetic therapies to cure sickle cell disease. The painful and potentially life-threatening genetic condition affects nearly 100,000 people in the United States, more than half of whom are enrolled in Medicaid, Medicare, or both (Centers for Medicaid and Medicare Services or CMS). Conventional treatments or common care for sickle cell can include the use of hydroxyurea and other medicines, as well as frequent blood transfusions. However, these treatments are largely temporary and focus on easing the symptoms of the disease, particularly severe pain episodes that are the hallmark of SCD. Bone marrow stem cell transplants, while curative, are limited by a lack of genetically well-matched donors and other challenges.

Launched in September 2018 by the National Heart, Lung, and Blood Institute (NHLBI), the #CureSC Initiative aimed to transform the lives of individuals living with SCD by moving the most promising gene therapies safely into clinical trials. These efforts culminated in the FDA approval of two gene therapies for sickle cell disease in December 2023. These gene therapies boost the production of fetal hemoglobin, a type of oxygen-carrying blood protein present at birth. When increased in adults with SCD, fetal hemoglobin reduces complications associated with the disease, including blockage of the blood vessels that can cause severe pain and organ damage, including stroke. Experts have long estimated that the cost of these therapies will be extremely high initially, but previous studies on the cost have been largely hypothetical.

Director Anirban Basu and Affiliate Faculty Dr. Scott Ramsey (Fred Hutch) co-led the CureSC Clinical and Economic Impact Analysis Consortium (CEIA - https://curesickle.org/content/clinical-economic-impact) for the National Heart, Lung, and Blood Institute of the National Institutes of Health. The goal of this Consortium was to help identify and evaluate the cost effectiveness of gene therapies for SCD. The Consortium was funded by the National Heart, Lung, and Blood Institute (NHLBI) through awards to the University of Washington (UW) and the Fred Hutchinson Cancer Center (FH). The investigators from both institutions received feedback and guidance from an expert panel consisting of individuals living with sickle cell disease, advocates, clinicians with expertise in pediatric and



adult sickle cell disease care and bone marrow transplant, as well as bioethicists, health economists, database analysts, and members of the payer community.

The final and most comprehensive paper produced by the CEIA consortium followed the U.S. Food and Drug Administration's approval of the first gene therapies for sickle cell disease in December. The work suggests that compared with the life-long costs of treating sickle cell disease (SCD), a potential one-time gene therapy treatment will significantly reduce morbidity, improve survival, and reduce the cost of care over patients' lifetimes compared to conventional care. Published in the Annals of Internal Medicine, the study shows the results of two independent computer simulation models to provide the most detailed estimates of upfront and downstream clinical effects, costs, and value-based prices of gene therapy treatments for SCD to date. Both models predicted tremendous health benefits for patients and spillover benefits for caregivers and society in terms of enabling patients to stay in the workforce. The two models differed in the price where gene therapy would be considered cost-effective. One model found that a price of $1 million is needed, while the other estimated a price below $2 million per patient would be cost-effective. The question of affordability to treat SCD with gene therapy is front-and-center following the U.S. Food and Drug Administration's approval of the first gene therapies in late 2023. Estimates place the current lifetime cost of medical care for SCD patients using conventional treatments at around $2 million.

(continued on next page)

"Our main results indicate that if the acquisition cost of a gene therapy for sickle cell disease is less than $2 million, the therapy will provide society with an acceptable value," said corresponding study author Anirban Basu, Ph.D.

In this study the researchers developed two independent cost-effectiveness simulation models to explore these real-world costs as well as benefits over a lifetime. They included the University of Washington Model for Economic Analysis of Sickle Cell Cure (UW-MEASURE) and the Fred Hutchinson Institute Sickle Cell Disease Outcomes Research and Economics Model (FH-HISCORE). Both models, which differ slightly in their design, analyzed insurance claims data on individuals with sickle cell disease who were enrolled in Medicaid, Medicare, or both (Centers for Medicaid and Medicare Services or CMS) between 2008 and 2016.

The researchers limited their analysis to CMS data from 4,762 people ages 12 to 38 years (50% female) who were eligible for gene therapy based on current patient selection criteria. Researchers adjusted both models to reflect an average life expectancy for the SCD patient population in general of 51 years, which is much shorter than the average life expectancy for Americans in general, which is about 77 years, according to the Centers for Disease Control and Prevention. The population eligible for gene therapy had more severe disease and an even shorter life expectancy than others with SCD, the researchers noted.

In addition to cost estimates, the models also analyzed the effectiveness of treatments with a focus on health impacts, such as quality of life and life expectancy; and nonmedical impacts, including loss of time and patient productivity, and the impact on the patient's caregiver and family. Both models were compared to conventional care across the average lifespan of an SCD patient.

The FH-HISCORE model found that any price tag for SCD gene therapy below $1 million is highly cost-effective, while the UW-MEASURE model found that any price below $2 million is highly cost-effective. Collectively, the models suggest that any price below $2 million, while high, could be worth it. But above that, they noted, the confidence that such therapy will deliver an acceptable value for money deteriorates quickly.

In addition to the cost benefits over conventional therapy, both models showed other important advantages to SCD gene therapy, including societal benefits. It could increase life expectancy by 17 years (17.4 for UW-MEASURE and 17 for FH-HISCORE). It would reduce the lifetime number of acute pain crises by 69 or 86 events, respectively, compared to common care. And it could increase prospects for the patient's long-term employment, as well as offer improvements in their caregiver's quality of life.

Aside from those perks, said Basu, the price of gene therapy could also get cheaper. "As costs of manufacturing gene therapy products become more streamlined and efficient, it could help bring down the prices of gene therapy," he said, noting that the exact cost of gene therapy for SCD also depends, in part, on future discussions among policymakers, insurers, and patients, who must balance budgets, affordability, and promote competition across multiple manufacturers of gene therapy.

Study co-author Scott Ramsey, M.D., Ph.D., a researcher, health economist, and director of the Hutchinson Institute for Cancer Outcomes Research at the Fred Hutch Cancer Center in Seattle, noted the importance of looking beyond healthcare costs to fully appreciate the value of SCD gene therapy.

"Ignoring the societal costs of treating sickle cell disease—factors such as impact on a patient's ability to work and caregiver burden—vastly underestimates the potential benefits of gene therapy, " he said. "This is ultimately about saving lives."

This work was funded by NHLBI's Cure Sickle Cell initiative under agreements OT3HL152448 and OT3HL151434. For a more complete funding disclosure and list of authors, please see the full research article.

STUDENT ACHIEVEMENTS

2024 RECIPIENTS OF THE LOUIS SR. AND MARILYN GARRISON ENDOWED PRIZE IN HEALTH POLICY AND ECONOMICS

This award is intended to recognize annually outstanding student research papers in the area of health policy and economics. Professor Emeritus Lou Garrison and his wife Fran established this fund in 2008 in fond memory of his parents Louis P. Garrison, Sr., and Marilyn J. Garrison, who were great believers in the value of education and science. This year we were pleased to present this award to two students. Congratulations, Spencer and Shiven!



Shiven Bhardwaj
PhD Student

PAPER TITLE: *Receipt of Routine Pediatric Vaccine in Community Pharmacy is Not Associated with Delayed or Missed Well-Child Visits*



Spencer Cheng
MS Student ('23)

PAPER TITLE: *Productivity loss by cancer stage in patients newly diagnosed with hepatocellular carcinoma: A claims database analysis*

RECIPIENT OF THE 2024 UW SCHOOL OF PHARMACY GRADUATE STUDENT LEADERSHIP AWARD



Ashley Tabah
PhD Student

This Spring, Ashley was selected as a recipient of the School of Pharmacy 2024 Graduate Student Leadership Award. The award decision is based on the consideration of outstanding leadership and service that positively impacted the School, UW and beyond.

Ashley Tabah is highly deserving of the 2024 UW School of Pharmacy Graduate Student Leadership Award. Since joining the PhD program in 2020, Ashley has demonstrated exceptional leadership, academic prowess, and dedication to her peers and field. Ashley's academic accomplishments are notable. She played a pivotal role in a research team, contributing to a grant application to the National Institute of Aging and subsequently securing her own grant to support further research. Her work has resulted in conference presentations, a peer-reviewed manuscript, and has laid the foundation for her dissertation, which focuses on critical policy issues for older adults.

During her time in the PhD program, Ashley has served as an elected student representative, curriculum committee member, social chair, and is currently on the search committee for the next Associate Dean for Research and Graduate Programs. In these roles, she has been an advocate for her peers, improved the graduate program, and fostered a strong community. Ashley's integrity, dedication, and genuine concern for others make her an outstanding candidate for this award. One of her nominators, Professor Ryan Hansen states, "It has been my experience that Ms. Tabah is one of those individuals who rarely if ever declines an opportunity to be a leader and serve." Fellow nominator, Professor Josh Carlson, echoes this sentiment and said, "In all of these roles, Ashley has exemplified the qualities of a leader: integrity, dedication, and a genuine concern for the well-being of others. She has consistently worked for the betterment of the CHOICE program and the UW School of Pharmacy, demonstrating an ability to balance her academic responsibilities with her leadership roles." Congratulations, Ashley, on this well-deserved honor!

FELIPE MONTANO-CAMPOS AWARDED LATINO CENTER FOR HEALTH FELLOWSHIP



Congratulations to Felipe Montano-Campos, a fourth-year PhD student at the CHOICE Institute. This year, Felipe was selected to serve as a Latino Center for Health Fellow. An interdisciplinary health sciences research center at the University of Washington, the Latino Center for Health is committed to improving the health and well-being of Latinx individuals, families, and communities in Washington state.

Felipe's current work focuses on modeling precision screening for cancer patients and examining how patient-doctor concordance in race/ethnicity and gender can enhance treatment and medical testing adherence, particularly among Latinx populations. Concurrently, his PhD dissertation intersects economics, decision science, and epidemiology, aiming to refine epidemiological models by integrating economic behaviors and principles, such as risk compensation and health-wealth trade-offs. This integration provides insights into how individuals adjust their behavior in response to perceived risks and the implications on policy outcomes. His research emphasizes how risk perception shapes choices about private preventive actions, highlighting the importance of public health policies that are effective and sensitive to the complexities of human behavior. Congratulations, Felipe!

CHOICE PHD STUDENT AWARDED ASCO CONQUER CANCER MERIT AWARD



This Fall, Sara Khor, PhD ('24) was awarded the the American Society of Clinical Oncology (ASCO) Conquer Cancer Merit Award for her team's study that examined the association between pre-diagnosis financial strain and later stage cancer diagnosis.

The team's research explores the relationship between pre-diagnosis financial strain (FS), measured through credit reports, and later stage cancer diagnosis. Analyzing data from cancer patients in Western Washington (2014-2018), the study reveals that FS is associated with later stage cancer diagnosis, which could lead to adverse cancer outcomes. Among breast cancer patients, this association can be largely explained by nonreceipt of mammography. Their findings suggest that even with insurance coverage, individual's financial stress may create an additional barrier to timely medical attention for suspicious cancer symptoms or routine screening, further exacerbating cancer disparities. This research highlights the need to develop strategies to improve access to care for these individuals.

In collaboration with her collaborators at the Hutchinson Institute for Cancer Outcomes Research, Veena Shankaran, Catherine Fedorenko, Kaiyue Yu, Zach Rivers, and Shannon Kestner, this study was also selected as a featured abstract in the Journal of Clinical Oncology.

Read the full publication here:

Khor S, Yu K, Fedorenko CR, Rivers Z, Kestner S, Shankaran V. Association between pre-diagnosis financial strain and later stage cancer representation. JCO Oncology Practice 2023 19:11_ suppl, 127. Published online November 2023.

RACHEL WITTENAUER RECEIVES 2024 SCHOOL OF PHARMACY OUTSTANDING DISSERTATION AWARD



Rachel Wittenauer, MPH, PhD ('24), was selected for the 2024 School of Pharmacy Outstanding Dissertation Award, reflecting the significant scientific, policy, and societal impact of her work. Rachel's dissertation, titled "Equitable access and reimbursement for pharmacy-based services: A case study of adult vaccinations," addresses critical issues in healthcare accessibility and equity.

Her research begins by defining and mapping pharmacy deserts in the U.S., identifying 15.8 million people living in areas without adequate pharmacy access. These populations often face multiple social determinants of health, such as lower educational attainment, racial/ethnic minority status, and lower health insurance coverage. This pioneering study, published in Health Affairs Scholar, provides the first localized estimates of pharmacy deserts nationwide and has informed state-level healthcare policy.

Rachel's work extends to evaluating the impact of pharmacy deserts on shingles vaccination rates, revealing that broader classifications of low pharmacy access correlate with lower vaccination completion rates. Her findings underscore the direct health consequences of limited community pharmacy access. Additionally, Rachel assessed the effects of state-level provider status legislation for pharmacists on reimbursements for vaccination visits, uncovering a significant gap between legal authority and practical implementation. This groundbreaking analysis offers a foundation for future research on the implications of provider status laws.

Rachel's research has already influenced policy. She presented her findings to the Washington State House Healthcare and Wellness Committee, contributing to the passing of legislation addressing pharmacy benefit managers. The Washington State Pharmacy Association utilized her study to support this bill, which was signed into law in March 2024.



RACHEL WAS PRESENTED THIS AWARD BY HER ADVISOR, PROFESSOR ANDY STERGACHIS, AT THE 2024 SCHOOL OF PHARMACY GRADUATE RECOGNITION CEREMONY. PHOTO CREDIT: GRADIMAGES

Rachel's leadership in her field has been recognized by the American Pharmacists Association, which invited her to a CDC-funded workshop in Washington, DC, to develop a tool for enhancing equitable access to vaccinations. As she prepares to join WHO's International Agency for Research on Cancer for postdoctoral work, Rachel is poised to continue making significant contributions to public health and policy.

Her dissertation's implications for patient health, policy solutions, and future research highlight Rachel's exceptional impact on healthcare access and equity, making her a deserving recipient of the Outstanding Dissertation Award.

AWARDS AND ACTIVITIES: **FACULTY**

ANIRBAN BASU TO SERVE ON THE ADVISORY COUNCIL FOR THE STATE OF ILLINOIS



Professor Anirban Basu has been appointed as a standing member to serve on the Advisory Council on Financing and Access to Sickle Cell Disease Treatment and Other High-Cost Drugs & Treatments for Governor JB Pritzker of Illinois. Professor Basu co-led the CureSickleCell Initiative's Cost and Economic Impact Analysis Consortium for NHLBI, culminating in the economic evaluation and value-based pricing of the recently approved gene therapies in sickle cell disease.

ANDY STERGACHIS NAMED AAAS FELLOW



Professor Andy Stergachis has been named an American Association for the Advancement of Science (AAAS) Fellow. He is among 502 newly elected fellows from around the world, who are recognized for their "scientifically and socially distinguished achievements" in science and engineering. A tradition dating back to 1874, election as an AAAS Fellow is a lifetime honor, and all fellows are expected to meet the commonly held standards of professional ethics and scientific integrity.

Professor Stergachis was recognized for his work to better monitor the safety of essential medicines and vaccines, especially in low- and middle-income countries. He directed a study assessing the safety of antimalarial drugs among pregnant people in sub-Saharan African nations and has been involved in several other initiatives to assess the safety of vaccines used during pregnancy. He researches the global burden of antimicrobial resistance and has strengthened pharmacy services in numerous countries. Dr. Stergachis is an elected member of the National Academies of Medicine, fellow of the American Pharmacists Association and fellow of the International Society for Pharmacoepidemiology.

JING LI SERVES AS PANELIST ON NATIONAL AGING SYMPOSIUM



Associate Professor Jing Li served as an invited panelist at a national symposium co-organized by Massachusetts Institute of Technology (MIT)'s AgeLab and American Association of Retired Persons (AARP), titled "Before the Diagnosis: Dementia's Early Financial Toll". The symposium was inspired by latest research including her first-authored publication in JAMA Neurology that shows substantial wealth decline among U.S. older adults before dementia onset.

For more information on this study: https://jamanetwork.com/journals/jamaneurology/article-abstract/2809716)

CLAYTON ENGLISH MODERATES PANEL ON MENTAL HEALTH CRISIS IN HEALTHCARE



In February, the University of Washington School of Pharmacy hosted a panel discussion on the mental health crisis facing our frontline healthcare workers as part of the annual Nudelman Endowed Lecture series. The event focused on strategies for mitigating the current mental health crisis and the burnout experienced by healthcare professionals in the aftermath of the COVID-19 pandemic.

The event was moderated by Dr. Clayton English, CHOICE Faculty and Assistant Professor at the school of pharmacy with expertise in psychiatric pharmacy. The event featured a distinguished panel of thought leaders in government, psychiatry, health services research, and other fields including Keynote addresses from US Representative Adam Smith and Washington State Representative Vandana Slatter. The panel engaged in an open dialogue about the ongoing mental health crisis, burnout among frontline workers, and the policy solutions being developed in Washington State and at the UW to help address these issues.

JENNIFER BACCI NAMED A 2024 NAM FELLOW



The National Academy of Medicine (NAM) has selected Associate Professor Jennifer Bacci as one of seven outstanding health professionals for the class of 2024 NAM Fellowships. The fellows were chosen based on their professional qualifications, reputations and accomplishments, and the relevance of their current field expertise to the work of the NAM and the National Academies of Sciences, Engineering, and Medicine. The fellows will collaborate with eminent researchers, policy experts, and clinicians from across the country during their two-year fellowship. They will also participate in initiatives convened by the National Academies to provide nonpartisan, scientific, and evidence-based guidance to policymakers, academic leaders, health care administrators, and the public.

DOUGLAS BARTHOLD APPOINTED TO WASHINGTON STATE PRESCRIPTION DRUG AFFORDABILITY BOARD



Research Associate Professor and CHOICE Faculty Fellow, Douglas Barthold, was appointed by Governor Jay Inslee to serve on the Washington State Prescription Drug Affordability Board (PDAB). Authorized under Senate Bill 5532, this five-member board is tasked with analyzing data to review price and cost growth of prescription drugs and set upper limits for drugs that may lead to significant excess cost. Dr. Barthold will bring expertise in health economics and outcomes research, with specific attention to analyses of healthcare costs, and to the effectiveness of state policies that affect out-of-pocket costs for patients. This appointment synergizes well with his research agenda, which aims to use health policies to improve chronic condition management and healthy aging.



Jennifer Bacci, PharmD, MPH, BCACP

- Named an American College of Clinical Pharmacy (ACCP) Fellow
- Appointed as Co-Director, Center for Pharmacy Practice Transformation
- Principal Investigator Community Pharmacy Foundation: Developing a model of care for medications for opioid use disorder at community pharmacies.
- Co-Investigator: American Heart Association. GROW-RURAL: A Global to Rural Innovation Network to Adapt Evidence-Based Cardiovascular Interventions to Context.



Josh Carlson, PhD, MPH

- Co-Director: Post-Doctoral Fellowship Program in Health Economics and Outcomes Research, Pfizer. 2024–2026



Aasthaa Bansal, PhD

- Completed Sabbatical as Visiting Scientist, Child Health Evaluative Sciences, SickKids Research Institute, Toronto, Canada.
- Co-Director: Post-Doctoral Fellowship Program in Health Economics and Outcomes Research, Genentech. 2024–2026
- Moderated 5th Annual CHOICE Institute Research symposium in October 2023.



Doug Barthold, PhD

- Promoted to Research Associate Professor, July 2024
- Appointed Co-Chair of the WPEC 2025 Scientific Committee.



Beth Devine, PhD, PharmD, MBA

- Guest Editor, Theme issues, “State of the Science of Pharmacogenomics Implementation in Healthcare Systems and Communities”. Frontiers in Pharmacology, a Frontiers journal.
- Podcast recording: UW Pharmacy Fridays. ¨Pharmacogenomics¨. Joint recording with Ken Thummel, PhD (UW Department of Pharmaceutics) and Erica Woodahl, PhD (University of Montana, School of Pharmacy).
- Lead presenter: ISPOR Atlanta Workshop: Addressing Key Challenges in the Setting of Tumor-Agnostic Drugs: Meeting an Unmet Need?¨ With Dan Ollendorf, Emma Mackay, Yilin Chen.



Clayton English, PharmD, BCPP, BCPS, BCGP

- Appointed to the American Association of Psychiatric Pharmacists Foundation (AAPPF) Board of Directors for a three-year term starting in January 2024.



Anirban Basu, PhD

- Invited speaker for the 38th Annual Rankin-Skatrud Memorial Lecture at the University of Wisconsin-Madison, Department of Population Health Sciences. Dr. Basu presented on Clinical Algorithmic Discrimination.



Lou Garrison, PhD

- First-place award with Meng Li from the Innovation and Value Initiative’s “Valuing Innovation Project”.
- Chair of new ISPOR Special Interest Group (SIG) on Global Access to Medical Innovation.
- Co-taught 2 short courses at ISPOR 2024 on A Health Economics Approach to US Value Frameworks, and Valuation of Innovative Drugs.
- Spoke in five issue panels at ISPOR 2024 on a variety of topics: global differential pricing, the Inflation Reduction Act, value-based arrangements, real option value, and the value of diagnostic information.



Ryan Hansen, PharmD, PhD

- Promoted to Professor, July 2024
- Continued appointment of Interim Chair, Department of Pharmacy.
- Co-Investigator: Community Pharmacy Foundation: Developing a model of care for medications for opioid use disorder at community pharmacies.
- Site PI: Immunoglobulin Replacement Therapy and Infectious Complications After CD19-Targeted CAR-T-Cell Therapy; 2023–2028; National Cancer Institute.
- Director: Post-Doctoral Fellowship Program in Health Economics and Outcomes Research, Bayer, 2024–2026.





Noémi Kreif, PhD

- Started appointment as Assistant Professor, January 2024.
- Co-Director: Post-Doctoral Fellowship Program in Health Economics and Outcomes Research, AbbVie. 2024–2026.
- Invited speaker for 5th Annual CHOICE Institute Research symposium, October 2023.
- Appointed Faculty Advisor, UW Student ISPOR Chapter



Andy Stergachis, PhD, BPharm

- Elected as American Association for the Advancement of Science (AAAS) Fellow 2024.
- Best Poster Prize. International Society for Pharmacovigilance Annual Meeting. Bali, Indonesia. November 2023.
- Co-Investigator: Creating a Region X center workplan to increase implementation of public health emergency preparedness and response (PHEPR) evidence-informed and evidence-based practice; US Centers for Disease Control.



Kyueun Lee, PhD

- Co-Director. Post-Doctoral Fellowship Program in Health Economics and Outcomes Research, AbbVie. 2024-2026
- Taught SMDM short courses in October 2023 and August 2024: "Infectious Disease Modeling Using Dynamic Compartmental Models in R"
- At SMDM 2023, organized a symposium on "Challenges and Opportunities in Global Health Research: A Panel Discussion"



Jing Li, MS, PhD

- Promoted to Associate Professor with Tenure, July 2024
- Associate Editor, npj Health Systems, a journal of Nature Portfolio.
- Invited presentation at the University of Pennsylvania's Leonard Davis Institute of Health Economics (Penn LDI) Research Seminar in January. "Social Preferences of U.S. Physicians: Experimental and Observational Data"



Sean D. Sullivan, BScPharm, MSc, PhD

- Co-Investigator: Informing Medicare Prescription Drug Price Negotiations: Integrating Net Prices and Comparative Effectiveness Assessments for the Initial Price Offer. The Commonwealth Fund. 2023-2024
- Director: Post-Doctoral Fellowship Program in Health Economics and Outcomes Research, Sanofi. 2024–2026
- Director: Post-Doctoral Fellowship Program in Health Economics and Outcomes Research, Novo Nordisk. 2024–2026"
- JMCP Award of Excellence – 2023 Paper of the Year "Drug Use and Spending Under a Formulary Informed by Cost-Eff ctiveness."
- 2024 PhRMA Foundation Challenge Award – Inflation Reduction Act
- 2024 Rho Chi National Honor Society Annual Lectureship



David Veenstra, PharmD, PhD

- Co-Director: Post-Doctoral Fellowship Program in Health Economics and Outcomes Research, Genentech. 2024–2026
- Appointed Co-Chair of the WPEC 2025 Scientific Committee.

Cate Lockhart, MS ('16), PharmD ('13), PhD ('16)

- Dr. Lockhart has been promoted to Chief Science Officer at the Academy of Managed Care Pharmacy (AMCP), combined with her role as Executive Director of the AMCP Biologics and Biosimilars Collective Intelligence Consortium (BBCIC). For the past six years, Dr. Lockhart has built BBCIC into a productive and widely respected non-profit research initiative with the mission to generate reliable real-world evidence evaluating the safety and effectiveness of biologic drugs to improve public health (www.bbcic.org). Her vision has led to the launch of the AMCP Research Institute, which offers a broadened research scope that includes the flagship research of BBCIC, but also allows AMCP to more broadly address the needs of managed care professionals with internal efforts. This new organizational initiative proactively contributes to the AMCP Strategic Priorities allowing us to directly approach research questions of particular interest to the profession of managed care pharmacy. With the trust and enthusiasm of AMCP staff nd Board of Directors, she is excited to continue building research programs that are important to managed care professionals.

Rafael Alfonso Cristancho, MD, MS, PhD ('13)

- Received a recognition as author of one of the top 10 most-cited papers published in The Clinical Respiratory Journal: Mepolizumab improves work productivity, activity limitation, symptoms, and rescue medication use in severe eosinophilic asthma (https://onlinelibrary.wiley.com/doi/full/10.1111/crj.13474)

Mark Bounthavong, PharmD, MPH, PhD ('18)

- Promoted to Professor of Clinical Pharmacy at UCSD Skaggs School of Pharmacy & Pharmaceutical Sciences
- Pilot grant from the National Institutes of Health, Grant UL1TR001442 - $30,000. Project title: "Quantifying health equity and financial risk protection for a nationwide naloxone distribution program among patients at risk for opioid-related overdoses: Application of the distributional cost-eff ctiveness analysis framework"
- Awardee, Distinguished Service Award - New Professionals Steering Committee Chair, ISPOR-The Professional Society for Health Economics and Outcomes Research
- Awardee, Distinguished Teaching Award for Academic Senate Members - University of California San Diego

Jonathan Campbell, MS, PhD ('07)

- 2024 PAA Distinguished Alumni - Pharmaceutical Sciences/Research Award
- 2024 St. Olaf College Athletics Hall of Fame Inductee (men's tennis)

Meng Li, PhD ('18)

- First Prize award recipient of the IVI Valuing Innovation Project (VIP) for her paper co-authored with Professor Emeritus Lou Garrison, titled, "Incorporating Real Option Value in Valuing Innovation: A Way Forward." This paper proposes a new approach to valuing innovation that incorporates real option value, arguing that this approach can better capture the value of innovative products and services that have the potential to improve health outcomes and reduce costs. Dr. Li's work in this area began with her dissertation research, when she was a PhD student. CHOICE members who contributed to this research include: Dave Veenstra, Anirban Basu, Zizi Elsisi, Woojung Lee, and Carrie Bennette

Jonathan Watanabe, PharmD, MS ('09), PhD ('12)

- UCI Institute for Precision Health Distinguished Lecture. "Data as a Public Good for the Public Good to Solve Immediate Real World Challenges" Inaugural speaker.
- 6th Annual California Institute for Regenerative Medicine Meeting (CIRM) Alpha Clinic Network Symposium. Invited Keynote Speaker. Health Economics, Outcomes Research, Affordability and Access Considerations for Gene and Cell-based Therapies.
- National Academies of Sciences, Engineering, and Medicine. Appointed as a Committee Member to "Processes to Evaluate the Safety and Efficacy of Drugs for Rare Diseases or Conditions in the United States and the European Union" Consensus Study Committee.
- National Academies of Sciences, Engineering, and Medicine. Appointed as Co-Chair for the Preparing the Future Workforce in Drug Research & Development Committee Workshop Proceedings: https://nap.nationalacademies.org/catalog/27755/preparing-the-future-workforce-in-drug-research-and-development-proceedings

Souvik Banerjee, MA, PhD, UW-Pfizer Postdoctoral Fellow 2014–2016

- Received the 2024 Research Excellence Award at the Indian Institute of Technology (IIT), Bombay, where Dr. Banerjee serves as an Assistant Professor in the Department of Economics. These awards were instituted by the IITB class of 1973, with the aim of cultivating a vibrant research culture at IIT Bombay.

CAREER UPDATES

Zach Baldwin, PharmD ('22), MS ('23)
Manager, HEOR Strategy, AbbVie

Jon Campbell, PhD ('07),
Chief Science Officer of the National Pharmaceutical Council (NPC)

Spencer Cheng, PharmD ('22), MS ('23)
Manager, Health Economics and Outcomes Research at Neurocrine Biosciences

Mitchell Higashi, PhD ('01)
Associate Chief Science Officer of ISPOR — The Professional Society for Health Economics and Outcomes Research

Boshen Jiao, PhD ('22)
Assistant Professor of Pharmaceutical and Health Economics at University of Southern California

Arvind Katta, PharmD, MS ('23)
Senior Manager, Bayer

Hyunwoo Kim, PharmD, MS ('23)
Manager, Health Economics and Outcomes Research at Neurocrine Biosciences

Jae Rok Kim, PharmD, MS ('23)
Manager, HEOR, AbbVie

Stella Ko, PharmD, MS ('23)
Associate Health Economist, Genentech

Woojung Lee, PharmD, PhD ('22)
Associate Director, Health Economics and Decision Modeling at ICER

Meng Li, PhD ('18)
Assistant Professor, Center for the Evaluation of Value and Risk in Health at Tufts Medical Center





Jean Malacan, PharmD, MS ('18)
Sr. Director, Global Value & Access Lead Hematology at BeiGene

Naomi Schwartz, PhD ('21)
Associate Director of Quantitative Science at BioMarin Pharmaceutical Inc.

Wei-Jhih Wang, MS, PhD ('20)
Senior Manager for HEOR/RWE at Moderna

Phoebe Wright, PharmD ('19)
Associate Director, Patient Centered Outcomes, Agios Pharmaceuticals

PUBLICATIONS

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2023–2024 IMPACT OF PRIVATE SUPPORT

3
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7
Faculty Received Endowed Salary Supplements

16
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18
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